

Serious Shea: Decision Support Tool

Submitted by TechnoServe

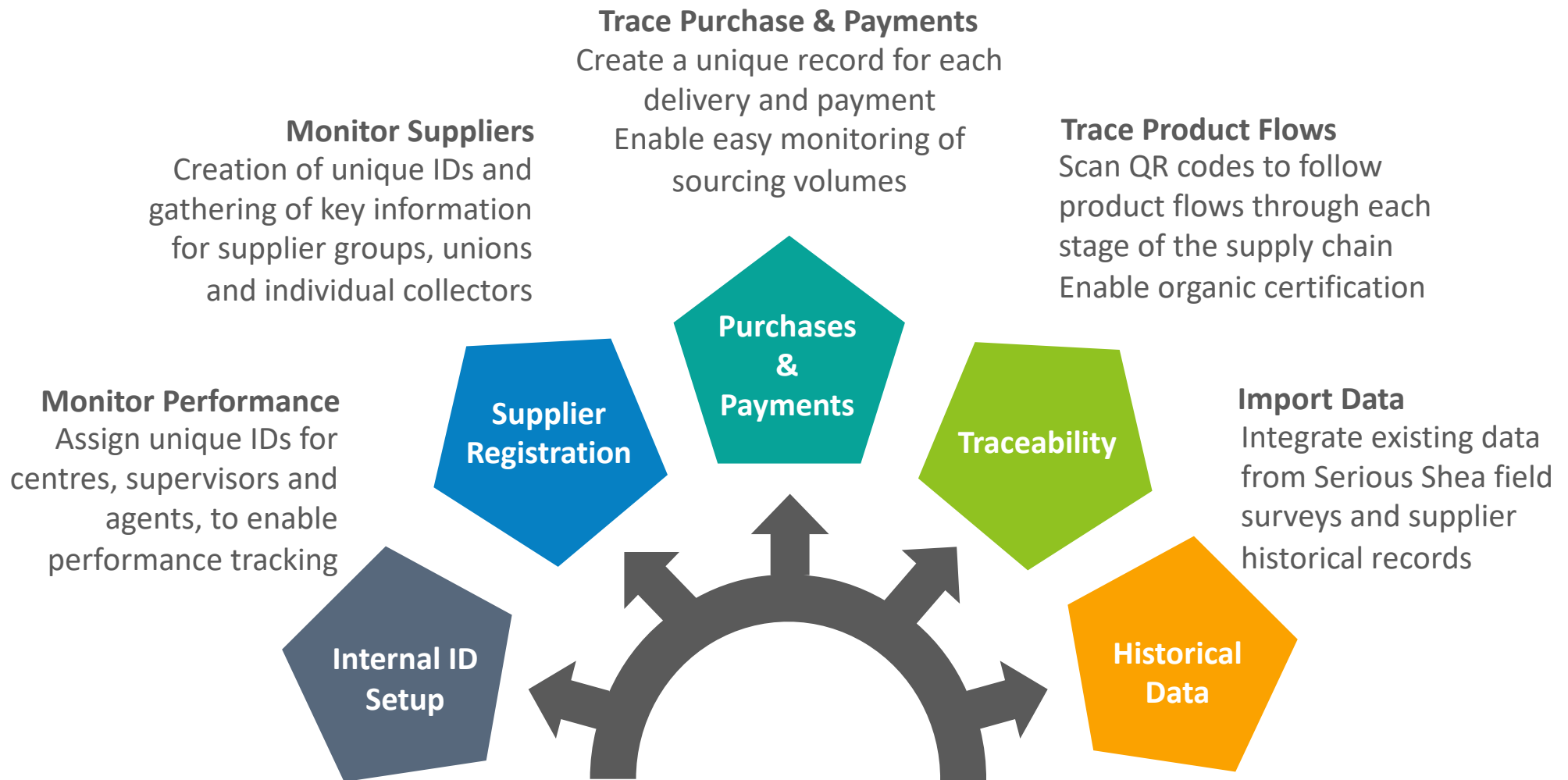
January 23, 2020

Overview

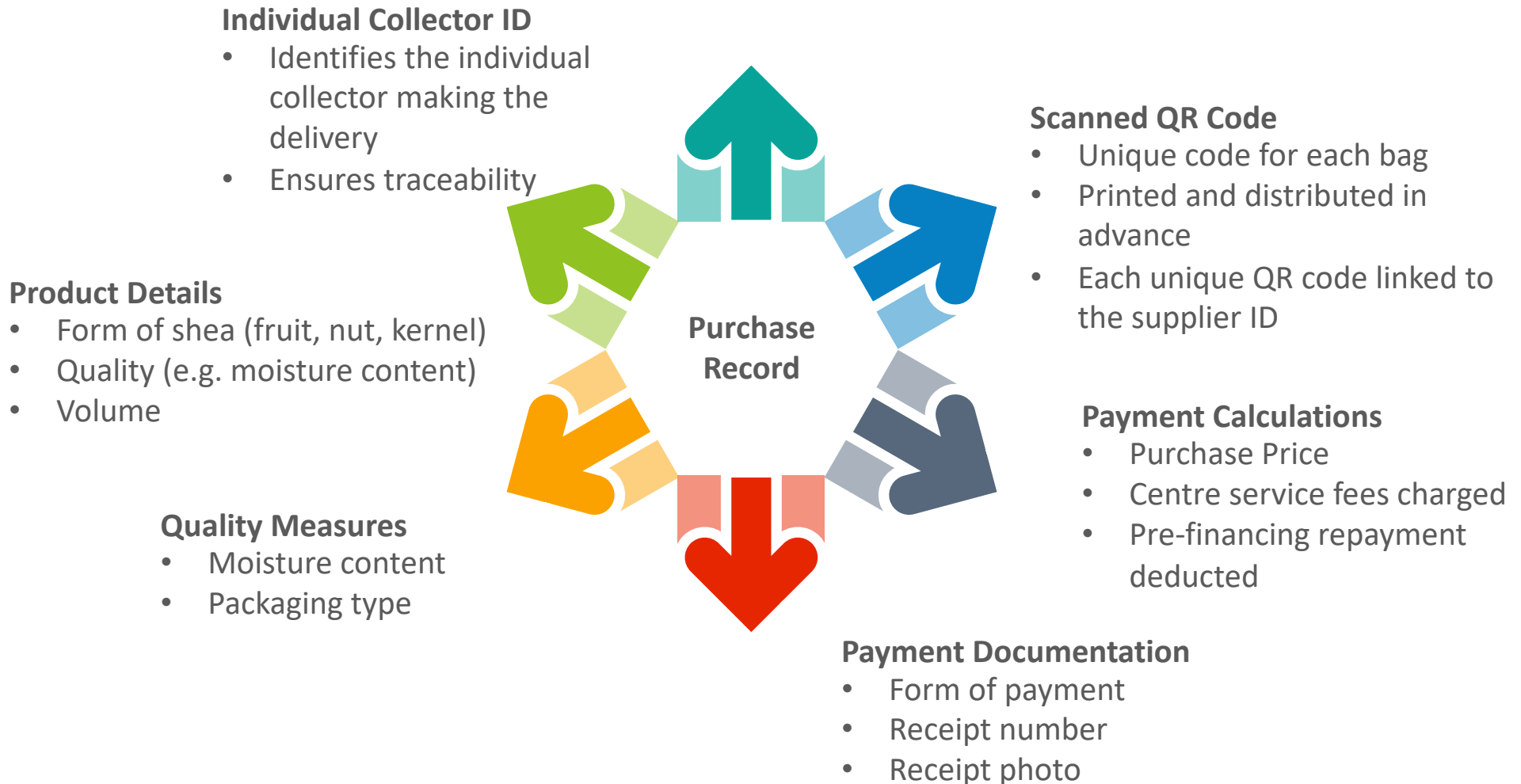
The DST will enable data-driven, evidence-based decisions regarding sourcing strategy through:



The DST will perform five major functions



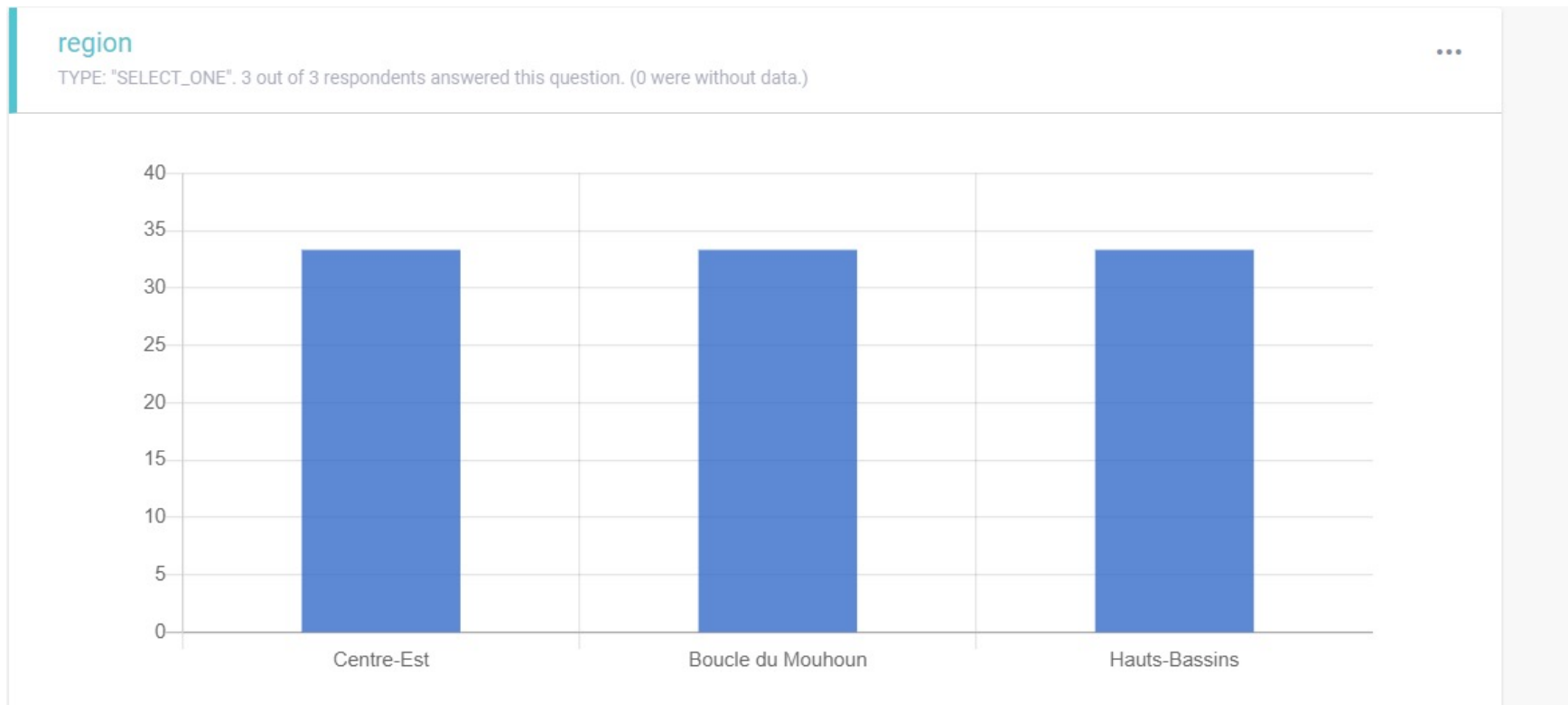
Data will be entered into the DST via mobile app to create each purchase record



The DST will provide visual dashboards

- **Volumes:** projections, purchases, processing
- **Payments:** aggregations by Centre, agent, Union/group, collector
- **Performance:** collectors registered, projections created, purchases made, volumes purchased by Centre, agent, Union/group, collector

Illustrative dashboard

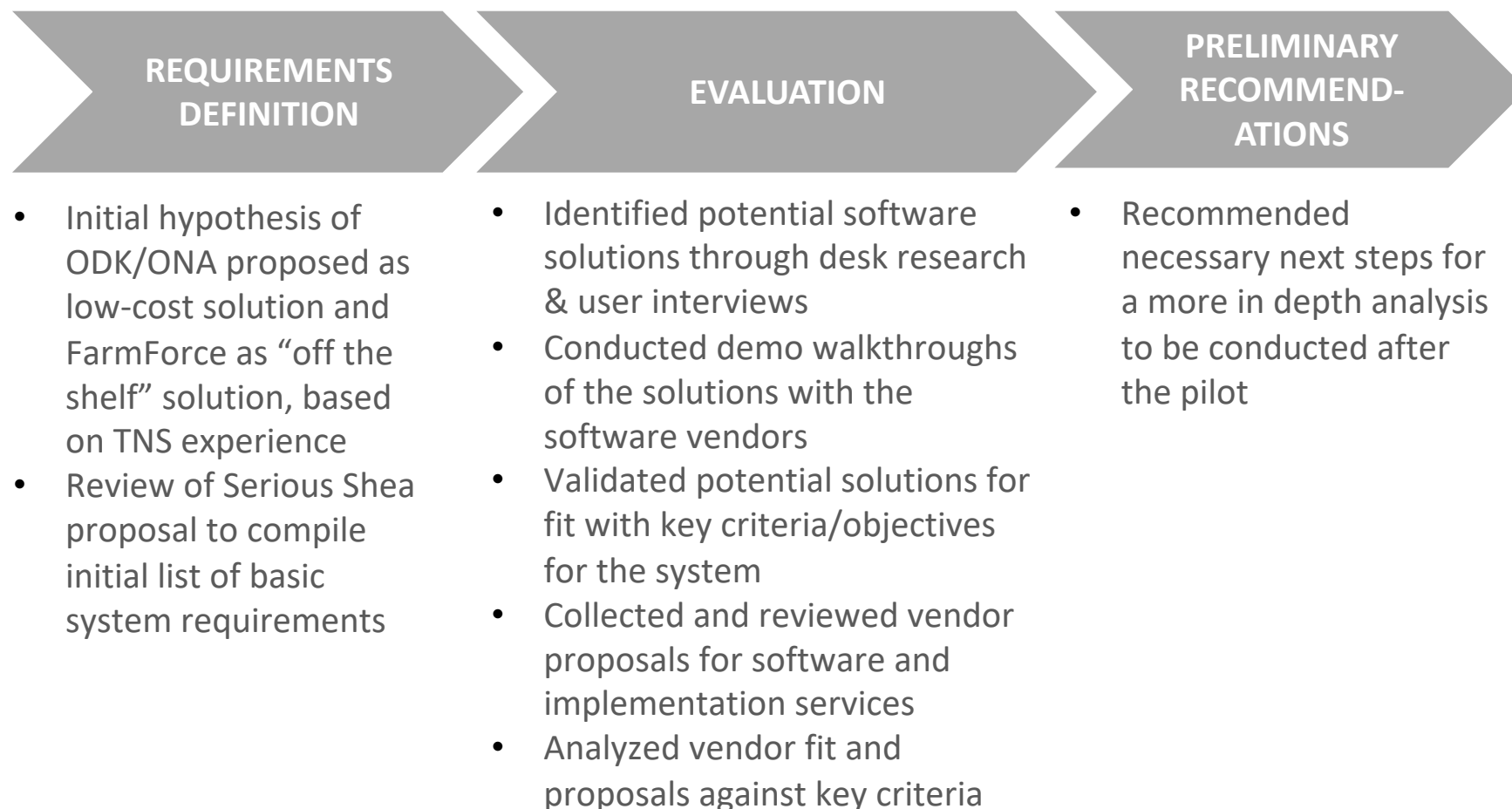


Methodology

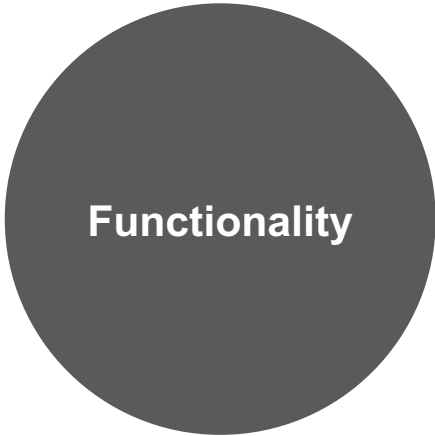
Tool Assessment

An assessment of tools has been conducted

- At project conception, TechnoServe made recommendations of software for use in the short- and medium-term
- A rapid assessment was done to shortlist solutions for the interim and long-term DST
- Following the rapid assessment, an interim tool was developed and a deeper analysis conducted for identified long-term DST solutions

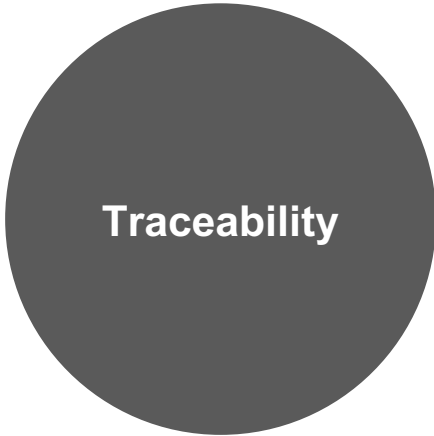


The assessment was done based on the following criteria:



Functionality

- Collector ID and demographic data management
- Geotagging (for mapping trees)
- Two way communication (SMS)
- Payments and supply tracking



Traceability

- Internet and IT Infrastructure dependence
- Need for & ease of customization
- Ease of use
- Support response time
- Cost

Different methods were used to evaluate 8 different tools

Platform	Method of evaluation		
	TechnoServe user experience	Demo walk-through	Desk research
FarmForce	✓	✓	✓
CropIn	✓	✓	✓
Insyte		✓	
SAP	✓		
CommCare	✓		
SurveyCTO	✓		✓
ONA	✓		✓
ODK	✓		✓

Tool Rapid Assessment

All tools were evaluated for functionality

Platform	Geotagging	GIS Maps (dashboards)	Traceability (barcodes)	2-way comm (SMS)	Mobile payment	Ease of scalability
FarmForce	✓	✓	✓	✓	✓	High
CropIn	✓	✓	✓	✓	✓	High
Insyt	✓	✓	✓*	✓	✓	Medium
SAP	✓		✓			High
CommCare				✓		Medium
SurveyCTO						High
ONA		✓		✓		High
ODK						High

Currently under development

Tool Rapid Assessment

All tools were evaluated for service & customer experience

Platform	Ease of use	Support response time	Support resources in French	Potential for customization
FarmForce	Easy	Hours	✓	✓
CropIn	Easy	Hours		✓
Insyt	Easy	Hours	✓	✓
SAP	Easy	Hours	Unknown	Unknown
CommCare	Moderate-advanced	Depends*		✓
SurveyCTO	Easy-moderate	Hours		
ONA	Easy	Hours		
ODK	Moderate	Varies		

* Guaranteed customer response time depends on license plan purchased, can be up to several days

Tool Rapid Assessment

All tools were evaluated for required investment

Platform	LOE for set-up	Training	Tablets/Smart phones	One-time set-up costs	Ongoing annual costs
FarmForce	13 days*	Beginner	Yes	\$12,500	~\$250 per user
CropIn	5 days	Beginner	Yes	\$10,000	\$200 per user + \$40k per year for mWarehouse
Insyt	5 days	Beginner	Yes	TBD	TBD
SAP	TBD	Moderate	Yes	TBD	TBD
CommCare	1-2 weeks	Moderate-advanced	Yes	Free	\$24,000+
SurveyCTO	5 days	Moderate	Yes	Free	\$2,300-\$2,500 per 5,000 forms
ONA	5 days	Moderate	Yes	Free	\$1,999 per year **
ODK	5 days	Moderate	Yes	Free	Free

* 5 days for set-up & data migration, 9 days for survey creation

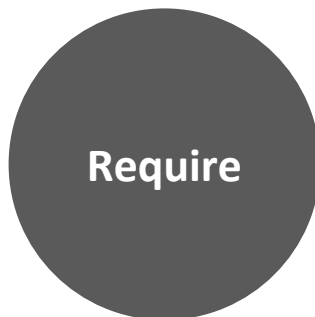
**\$1,999/yr for 50 surveys, 25,000 form submissions

The assessment found some commonalities among all 8 tools evaluated

All evaluated software platforms:



- App & web-based platform with offline access
- App functionality in areas with weak network access
- Unlimited data storage capacity



- Some level of user training
- Investment in mobile devices for field agents¹
- Investment in a PC for web app²

¹ Mid-range smartphones/tablet

² except Insyt, which offers paid data collection service, and therefore wouldn't require field agent devices)

The assessment resulted in a set of findings:

We found that free software, (ODK), has many weaknesses:

- X Lacks GIS mapping
- X Does not offer traceability
- X No integration of mobile payment functionalities

We found that the low-cost software (CommCare, SurveyCTO, ONA) have many weaknesses:

- X Offer free, but fairly labor intensive setup
- X Do not offer traceability
- X No integration of mobile payment functionalities
- X Do not offer GIS mapping (except ONA)
- X Cost highly variable
 - ONA: \$2k (<8,333 collectors)
 - CommCare: \$3k (<125 field agents)
 - SurveyCTO: \$6.9-7.5k (<5,000 collectors)

We found that off-the shelf software (FarmForce, CropIn, Insyt, SAP) also have weaknesses:

- X Require pricey one-time setup costs of between \$10-12.5k
- X Some (Insyt & SAP) were unresponsive to requests
- X Some (CropIn) allow only the provider to customize surveys & reports

At the same time, we found that off-the-shelf software offer many advantages:

- ✓ All offer practically no functionality constraints; anything can be developed
- ✓ Some (Insyt) offer collector debit cards that can be used for traceability and payments
- ✓ Some (SAP) already have tested solutions for shea in Ghana

The assessment generated conclusions that shaped the remaining research and work on the DST

Summary of key rapid assessment conclusions

- 1 ODK can be eliminated, based on functionality
- 2 SurveyCTO can be eliminated, based on cost
- 3 ONA and CommCare remain viable low-cost options
- 4 FarmForce and CropIn appear to be the strongest/most responsive off-the-shelf solutions, but Insyt and SAP should not be ruled out until we have more information

Following the rapid assessment, a list of final requirements for the long-term DST solution has been compiled

The long-term DST solution requires the following functionalities:

- Cloud-based software
- Optimized for mobile and desktop
- Works offline
- Stores and manages data on the women's groups
- Tracks movement of shea (bar codes, supplier ID), payments to the women's groups, and butter consignments
- Integrates GIS maps
- Compiles data collected from Serious Shea project components 1, 2 and 3

As a result of the rapid assessment, a new free software was identified for the interim and a deeper analysis for long-term solutions has been conducted

Interim Tool

- Based on the need for a simple, low-cost tool for immediate use during the pilot phase, TechnoServe recommends KoBoToolbox (ODK based)
- KoBoToolbox is a free, open source solution with certain data limitations

Longer-Term Solution

- In anticipation of evolving needs as Serious Shea scales, TechnoServe negotiated proposals from software providers for a comprehensive software solution that will address all limitations of the interim tool, once Serious Shea is ready to invest in a more sophisticated system
- Software providers pursued included FarmForce, CropIn, CommCare

Interim Tool

To balance systems requirements with budgetary constraints, TechnoServe selected KoBoToolbox as simple and free interim tool

Characteristics of KoBoToolbox:

- A free ODK-based data collection and management tool with online and offline capabilities
- Comprises a robust form builder tool for developing and designing ODK compatible forms within a simplified use and drag interface
- Offers support for complex forms using XLSF forms

As a free solution, KoBoToolbox presents certain data limitations

- The system can capture new data entered via data entry forms, but there is no way to do a mass upload or migration of existing data.
- Data migration may be possible to do in the future, but would require dedicated development work (and no guarantee it would work, giving this is all open source)
- There is no off-the-shelf way to integrate mobile payments
- Dashboard functionality is limited, and dashboards can be a bit static

However, work arounds have been identified for these limitations

- QGIS files will be saved in the Serious Shea dropbox
- Survey results and data from all partners has been collated and saved in a [googledrive folder](#)
- GPS coordinates will be captured via KoBoToolbox during field work
- Existing data will be manually entered into the system via individual data forms by ImpactAgri interns, prior to or during the pilot launch
- Payment information will be manually entered via the data forms
- Dashboards can be extended by investing in compatible tools like PowerBI

Required Resources

A dedicated manager and some low-cost equipment will be required to establish and maintain the interim DST

Human Resources:

- A full-time DST manager to drive setup, testing, and oversight
- This person should have experience in trouble shooting open source applications (as there is no dedicated support service for ODK or Kobo Toolbox) as well as developing custom dashboards & reports and analysis

Equipment

- Laptop for the DST manager (Operating System: Windows 8.1, Processor Type: Core i5(7th gen) processor, Memory: 8 GB RAM, Hard Drive: 250 GB, Graphics Card: 512MB video memory, Vendors: Lenovo, HP, Dell)
- Android smartphone or tablet for each user, including each Field Agent and Group Purchasing Agent
- Printer for QR codes (any available printer, however reinforced paper or lamination is recommended to ensure that barcodes stay intact and attached throughout the warehousing and transport process)
- Aquaboy moisture reader for each Field Agent and Group Purchasing Agent

The interim tool is live and ready to be used

TechnoServe has created several user guides for the Interim DST, available in annex:

Getting Started with KoboCollect

- Download and configure KoBoCollect, the Android App for KoboToolbox
- Links to basic KoboToolbox guides available online

Getting Started with KoboToolbox

- Creating data entry forms

Getting Started with QR Codes

- Accessing and using the QR code generator
- Downloading and using the QR code scanner App

Viewing Form Submissions on KoboToolbox

- Review data submissions for a selected data entry form
- Links to KoboToolbox guides on how to edit data forms in Excel, edit or delete individual data submissions, download data, manage permissions, add create login credentials for users

Creating Reports on KoboToolbox

- Create custom reports, including customized graphics
- Reuse previously generated custom reports

To gain access, download the KoBoCollect app and the surveys to an android device

- Use the guides to install the app and download the surveys on an android device or follow [these instructions](#) on how to collect and submit data with a non-android device
- KoBoCollect is an android app, which can be downloaded via playstore on any android device
- KoBoToolbox is the server backend, which can be viewed using any device (laptop, mobile phone, etc.)
- Use credentials to download the surveys to an android device:

<https://kobo.humanitarianresponse.info/>

username: ss_burkina

password: ss_burkina

All anticipated forms for the pilot have been created

FORMS FOR INITIAL SYSTEM SETUP

Center Registration	Establish Center ID numbers that can be linked to all related Union, Group and Collector records, including purchases. Map Centers.
Supervisor Registration	Establish Supervisor employee ID numbers. Link field agents' purchases to their associated supervisor, so volumes by supervisor can be tracked and compared.
Agent Registration	Establish buying agent employee ID numbers. Link purchases to agent to ensure transparency and allow for easy tracking of volumes/performance by agent.

FORMS FOR ONGOING FIELD ACTIVITY DATA ENTRY

Union Registration	Establish Union ID numbers that can be linked to all related group and collector records, including purchases. Map Union offices.
Group Registration	Establish Group ID numbers that can be linked to all related collector records, including purchases. Map Group offices.
Collector Registration	Establish Collector ID numbers for traceability. Link each purchase to the delivering collector. Allow identification of top performing collectors.
Volume Projections	Allow entry of volume projections for unions, groups or collectors, for aggregation of volume projections for various forms of shea.
Prefinancing	Allow for entry of credit agreements to unions, groups or collectors
Purchases & payments	Ensure traceability by capturing the details of each individual purchase. Scan QR codes. Keep records of each payment.

Data Forms

FORMS FOR MANUAL ENTRY OF EXISTING STUDY DATA

Union Volume History	Allow for manual data entry of data collected through TechnoServe's social assessment
Group Volume History	Allow for manual data entry of data collected through TechnoServe's social assessment
Collector History	Allow for manual data entry of data collected through TechnoServe's social assessment

Long-Term Solution

A deeper analysis of long-term software solutions was conducted

The following steps were taken:

- Re-establish contact with five software companies identified during the rapid assessment: FarmForce, CropIn, SAP, Insyst, CommCare
- Provide more detailed information on pilot and scale-up operations and required structure and functionality of the systems
- Request updated technical and financial proposals
- Re-evaluate functionalities based on defined system requirements
- Evaluate cost proposals



Recommendations:

- SAP & Insyst can be ruled out due to unresponsiveness and poor customer service
- CommCare can be ruled out due to lack of traceability function
- Both FarmForce and CropIn offer all the system requirements for the long-term DST and should be prioritized as the top two potential solutions

All tools were evaluated for functionality

	FarmForce	CropIn	CommCare
Geotagging	✓	✓	
GIS Maps (Dashboards)	✓	✓	
Traceability (Barcodes)	✓	Add-on feature	
Two Way Comm. (SMS)	✓	✓	✓
Mobile Payment Integration	Add-on feature	Add-on feature	
Ease of Scalability (Medium – High)	High	High	Medium

All tools were evaluated for customer service

	FarmForce	CropIn	CommCare
Ease of use	Easy	Easy	Moderate-Advanced
Support Response Time	Not specified	Emergency Request - 8 Hours Critical Request - 24 Hours Normal Request - 48-72 Hours	1 business day
Application available in French	✓	✓	✓
Customer service & client support in French	✓		
Customer Support Limits	3 free hours of support calls per month, \$40 per hour or \$320 per day, for excess support	Unlimited	Depends on plan

Required Investment

All tools were evaluated for required investment

	FarmForce	CropIn	CommCare
LOE Required for Setup	Setup & Data Migration: 5 days, Survey Creation: 8 days	5 days	1-2 weeks, depends on technical skill of user
Training Required	Beginner	Beginner	Moderate - Advanced
Tablets/ Smart phones needed?	Yes	Yes	Yes
Additional fees for custom dashboards or ad hoc support	\$350 a day for custom reports, \$400-500 a day for training of new users	None – all included	Custom dashboards not offered
One-Time Set Up Costs	\$12,500	\$10,000	No set up fee once a license is purchased
Ongoing Annual Costs/Subscription Fees	~\$250 per user*	\$200 per user	\$12,000 for 500 users (Advanced Plan)

*FarmForce has not provided a fixed rate per user, but has estimated that for 1,000 users, they would charge an annual amount of \$250,000

Cost Proposals: Assumptions

Providers were given the following assumptions on which to base their cost proposals

- Y0 is the pilot year, in which Serious Shea will use the free ODK/KoBoToolbox system
- Y1-Y3 are scaling years, cost proposals were provided for Y1

DST Users	Y0	Y1	Y2	Y3
Senior Managers	10	10	10	10
Centre Staff	20	60	175	350
Field Agents	65	201	601	1,202
Group Purchasing Managers	385	1,202	3,606	7,212
Total	480	1,473	4,392	8,774
Shea Collectors	11,539	36,058	108,174	216,347

- This model assumes that Group Purchasing Managers and Field Agents are the primary mobile app users
- Group Purchasing Managers use the DST to create payment records and track payments, to track collection of shea products from their members, and to register individual collectors
- Field Agents use the DST to register groups and unions, track movement and payment of shea products from group to collection centre, to processing centre
- Centre staff at the collection centers track details of bags received and processed into nut and kernel
- Processing centre staff track bags of kernel received and processed into butter

Cost Proposals: Overview

FarmForce

- Total Setup: \$52,500
- Annual Subscription: \$250,000 for 1,000 users
- **Estimated Total for Year 1: \$302,000**

CropIn

- Total Setup: \$10,000
- Annual Subscription: \$40,000 for mWarehouse + \$200,000 for 1,000 users
- **Estimated Total for Year 1: \$250,000**

CommCare

Advanced Plan

- Total Setup: \$0
- Annual Subscription: \$24,000 for 1,000 users
- **Estimated Total for Year 1: \$24,000***

*CommCare does not allow custom dashboards/visualizations, therefore, Serious Shea would need to invest in PowerBI, which is \$10 a month per user. This would add an additional \$120 to the Year 1 cost, assuming only the DST Manager has a PowerBI license. The enterprise plan, would require separate negotiation.

Total Cost

Setup

- Standard setup activities (training, configuration) - \$12,500
- Data Migration- \$10,000
- Custom Developments- \$30,000

Estimated Subscription Fees

- 1,000 users - \$250,000 per year
- 4,000 users - \$500,000 per year

Data Migration

- \$10,000: includes manual uploading of the following historical data for 36,000 collectors:
 - Enregistrement du centre - Harvest warehouse
 - Enregistrement de superviseur - Mobile user or group leader
 - Enregistrement d'agent - Purchasing agent - Mobile user
 - Enregistrement d'Union - Coop. Made up of groups. - Large group.
 - Enregistrement de groupement - Collector group - standard setup
 - Enregistrement de collectrice - Female collector
 - Prefinancement – without photo of the credit agreement.

Custom Developments

- Volume projections: FarmForce only has a projection function for one harvest type - fruit volume or kernel volume. Anything else would be a custom software development which would involve extra fees.
- \$30,000 to upload the following, more complex historical data:
 - Achat – Develop custom scripts for fees deducted and Photo of purchase receipts
 - Historique de volume d'Union
 - Historique de volume de Groupe
 - Historique volume collectrice

Cost Proposal: CropIn

Total Cost

Setup

- Standard setup activities (training, configuration) - \$10,000

Estimated Subscription Fees

- 1,000 users – \$200,000 per year
- mWarehouse license - \$40,000 per year

Data Migration

- Data migration to be a manual process with no extra cost, but amount of time investment is unclear. CropIn would provide an upload template

Custom Developments

- No need for custom developments identified by the vendor at this time

Cost Proposal: CommCare Available Plans

CommCare offers a selection of standard packages

- TechnoServe recommends either the **advance** or **enterprise** options based on Serious Shea's needs
- The standard and pro plans are not viable options as they do not allow integration with PowerBi, therefore dashboards would not be feasible

Standard	Pro	Advanced	Enterprise
PAYING ANNUALLY	PAYING ANNUALLY	PAYING ANNUALLY	PAYING ANNUALLY
\$250 monthly	\$500 monthly	\$1000 monthly	Custom
For programs with one-time data collection needs, simple case management workflows, and basic M&E requirements.		For programs with distributed field staff, facility-based workflows, and advanced security needs. Also for M&E teams integrating data with 3rd party analytics.	For organizations that need a sustainable path to scale mobile data collection and service delivery across multiple teams, programs, or countries.
READ MORE	READ MORE	READ MORE	READ MORE

Source & for more information on the differences between plans is available at <https://www.dimagi.com/commcare/pricing/> - enterprise packaging is negotiated individually based on the requirements of the system Full details

Total cost (dependent on plan purchased)

- No set up costs, only a license fee for a plan
- No data migration costs
- Unlimited form submissions
- Unlimited data storage
- All custom developments are based on negotiations with Dimagi

Estimated Subscription Fees

- Standard Plan – 125 mobile users at \$3,000 per year \$2 per additional mobile user per month
- Pro Plan – 250 mobile users at \$6,000 per year, \$2 per additional mobile user per month
- Advanced Plan – 500 mobile users at \$12,000 per year, \$2 per additional mobile user per month
- Enterprise Plan – unlimited users, custom price based on negotiation

Conclusions

FarmForce and CropIn are sophisticated softwares that should satisfy Serious Shea's needs, with some differences

- FarmForce is more expensive, but offers better services to francophone countries
- CropIn provides more extensive customer service & customization without additional fees

	FarmForce	CropIn	
Total Estimated Cost for Year 1	\$302,000	\$250,000	
Customer service & client support in French	✓		● If the DST Manager is fluent in English, CropIn's lack of francophone service reps should not be an issue
Support Response Time	Not specified	Emergency - 8 Hours Critical - 24 Hours Normal - 48-72 Hours	● Important to negotiate quick response time with FarmForce
Customer Support Limits	3 free hours of support calls per month, \$40 per hour or \$320 per day, for excess support	Unlimited	● If limited support and customization is necessary, FarmForce charging extra should not be a major issue
Additional fees for custom dashboards or ad hoc support	\$350 a day for custom reports, \$400-500 a day for training of new users	None – all included	●

Once ready to invest in the long-term DST, Serious Shea should evaluate & decide between FarmForce and CropIn

- Thorough analysis confirms that both FarmForce and CropIn are sophisticated software that cover all known required functionalities of the long-term DST
- Based on initial financial proposals, FarmForce appears more expensive than CropIn (higher price per user, slightly higher set-up costs, plus fee (~\$10k) for data migration)
- Detailed demos will allow the future DST manager to decide upon a software based on his/her preferred interface

Recommended next steps, in chronological order:

- 1 Confirm final requirements (slide 18) for DST remain true
- 2 Determine available budget
- 3 Conduct demos of both stock applications
- 4 Based on demos, identify required customizations
- 5 Ask vendors for a timeline and cost to build those customizations
- 6 Provide vendors with updated user numbers (slide 35) & data migration needs
- 7 Request and review updated cost proposal
- 8 Request and review proposed service-level agreement, negotiate terms & budget
- 9 Select & engage vendor for setup, customization, and piloting

Next Steps

Serious Shea will need to request updated cost proposals based on the evolution of the pilot in order to decide on a scaling solution

- No provider can give an accurate quote over six months in advance for integration with a system that does not yet exist
- The number of users will be a primary cost driver in any system, so refining the projected number of collectors and system users (especially Field Agents and Group Purchasing Managers Agents) will be critical
- TechnoServe has provided general guidance on the indicative price quotes provided to date, but Serious Shea will need to go back to the providers for updated quotes, based on pilot results

Provider Contacts

- FarmForce: Spencer Morley, spencer.morley@farmforce.com; Laura Johnson Blair, laura.johnson.blair@farmforce.com and Faith Kamenchu, faith.kamenchu@farmforce.com
- CropIn: Brighu Ravi Bhattrahalli, brighu@cropin.com; Varun Bhanot, varun.bhanot@cropin.com
- CommCare: Dev More, dmore@dimagi.com or <https://www.dimagi.com/commcare/>

Several points for potential cost savings should be explored with the providers

- The cost proposals are high because until the operational model is validated and the user numbers are finalized for the subsequent years, it is impossible to identify the points on which Serious Shea can negotiate down the setup and licensing fees.
- Since both FarmForce and CropIn base their prices on number of mobile users, Serious Shea could re-evaluate the role of the group purchase manager* and consider a model where field agents are the primary mobile app users. In this scenario, group purchase managers manage data via an excel spreadsheet, put printed QR codes on shea bags, and the field agents scan the QR codes, create the payment record, and validate the data from the excel spreadsheet. In this model, however, it would be easier to falsify records, therefore, Serious Shea would need to create a system that has appropriate checks and balances to manage this risk.
- Sometimes a single user can be spread across multiple agents for data collection. If this can be done, it could significantly reduce annual licensing fees.
- Current estimates on investments required for historical data integration are high for both FarmForce (estimated at \$10k) and CropIn (a manual process, likely time consuming). Data migration could be significantly reduced. Over the course of the pilot, Serious Shea should look critically at what historical data are actually being used, and par back the manual migration to only essential data.
- Custom development costs are also very high and could be reduced significantly. Over the course of the pilot, Serious Shea should look critically at what features are essential, vs. “nice to have” and base final design decisions on cost-benefit analysis of various features.